

DATA ANALYST ASSIGNMENT ONE:

CREATE AN RCTFCE FRAMEWORK PROMPT

ROLE:

You are a senior retail data analyst with over 8 years of experience in store optimization, product performance analysis, and customer retention.

CONTEXT:

A 30-store chain's performance has slid over 18 months. We have ~50K records with store, date, product, category, revenue, units, customer, loyalty points, and region. Investigate the cause and identify opportunities for improvement

TASK:

List the top 5 analytical questions (SMART) that link stores, products, and customers to the performance issues. Also specify one key metric or KPI relevant to each question (e.g., same-store sales growth, category churn rate, repeat purchase rate). These questions should yield information on which stores require closure or intervention, which products should be promoted, and which customers should be targeted for re-engagement.

FORMAT:

Provide a numbered list. For each question, include:

- (a) The question,
- (b) Why it matters to the CEO
- (c) The KPI or metric name,
- (d) The columns needed,
- (e) The type of insight expected.
- (f) The possible management action

CONSTRAINTS:

Questions must be specific and measurable. Avoid statistical jargon. Use only information available in the dataset. Focus on stores, products, and customers. Ensure the questions are understandable to executives.

EXAMPLE:

What is the monthly revenue trend for each store during the last six months?

- Matters because: stores with sustained declines may require intervention or closure.
- KPI: % revenue change.
- Columns: store_id, revenue, transaction_date.
- Expected insight: identification of consistently underperforming stores.
- Possible action: Implement improvement initiatives or consider store consolidation

STAKEHOLDER BRIEF

Prepared for: Chief Executive Officer (CEO)
Prepared by: Retail Analytics Team
Subject: Investigation into Declining Performance and Opportunities for Improvement

1. Situation Summary

The company's 30-store network has experienced declining performance over the past 18 months. To identify the causes and opportunities for improvement, the Analytics Team will analyze approximately 50,000 records covering store performance, product sales, customer transactions, and loyalty programme activity. The analysis aims to support decisions on store intervention or closure, product promotion, and customer re-engagement to improve profitability and long-term growth.

2. Three Key Business Questions

Area	Key Business Question	KPI	Decisions Informed	Data Variables
Stores	Which stores have experienced the greatest revenue decline over the last six months?	Same-Store Sales Growth (%)	Operational intervention and Store closure evaluation	store_id, revenue, transaction_date, region
Products	Which products and categories are losing sales, and which contribute most to revenue?	Category Sales Growth Rate	Promote high-performing products; Category rationalization	product_id, product_name, category, revenue, units_sold, transaction_date
Customers	Which previously active customers have reduced or stopped purchasing?	Repeat Purchase Rate	Customer re-engagement campaigns; Loyalty programme enhancement	customer_id, transaction_date, revenue, loyalty_points

3. Description of Available Data

Approximately **50,000 records** spanning **18 months** across **30 stores** are available. Variables available include: **store_id, transaction_date, product_id, product_name, category, revenue, units_sold, customer_id, loyalty_points, and region.**

Expected Outcome: The analysis is expected to identify underperforming stores, reveal products requiring strategic attention, and highlight high-value customers for re-engagement, thereby providing management with data-driven recommendations for improving profitability and long-term growth.

Purpose: To provide the CEO with actionable insights for improving the performance and profitability of the 30-store retail chain.

Reflection

I structured the prompt using the RCTFCE framework to ensure that the AI-generated responses were clear, actionable, and aligned with the CEO's information needs. The prompt emphasizes stores, products, and customers because these areas are critical to understanding the causes of declining performance and identifying growth opportunities. Including specific formatting requirements and examples helped ensure that the output was executive-friendly and suitable for decision-making.